

Personality and Behavior Changes

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Healthy people differ significantly in their overall personality, mood, and behavior. Each person also varies from day to day, depending on the circumstances. However, a sudden, major change in personality and/or behavior, particularly one that is not related to an obvious event (such as taking a medication or losing a loved one), raises concern for a mental health issue.

(See also [Overview of Mental Illness](#).)

Sudden changes in personality and behavior can be roughly categorized as involving one of the following types of symptoms:

- Confusion or delirium
- Delusions
- Disorganized speech or behavior
- Hallucinations
- Mood extremes (such as depression or mania)

These categories are not specific mental illnesses. They are just one way doctors organize different types of abnormal thought, speech, and behavior. These changes in personality and behavior can be caused by general medical conditions or mental health issues.

People may have more than 1 type of change. For example, people with confusion due to a medication interaction sometimes have hallucinations, and people with mood extremes may have delusions.

Confusion and delirium

Confusion and [delirium](#) refer to a disturbance of consciousness. That is, people are less aware of their environment and, depending on the cause, may be excessively

agitated and belligerent or drowsy and sluggish. Some people alternate between being less alert and being overly alert. Their thinking appears cloudy and slow or inappropriate. They have trouble focusing on simple questions and are slow to respond. Speech may be slurred. Often, people do not know what day it is, and they may not be able to say where they are. Some cannot give their name.

Delirium often results from a serious, newly developed general medical problem or a reaction to a medication, especially in older adults. People who have delirium need immediate medical attention. If the cause of delirium is identified and corrected quickly, delirium often resolves.

Delusions

Delusions are fixed false beliefs that people hold despite evidence to the contrary. Some delusions are based on a misinterpretation of actual perceptions and experiences. For example, people may feel persecuted, thinking that a person behind them on the street is following them or that an ordinary accident is purposeful sabotage. Other people think that song lyrics or newspaper articles contain messages that refer specifically to them (called a delusion of reference).

Some beliefs seem more plausible and can be difficult to identify as delusions because they could occur or have occurred in real life. For example, people occasionally are followed by government investigators or have their work sabotaged by coworkers. In such cases, a belief can be identified as a delusion by how strongly people hold the belief despite evidence to the contrary.

Other delusions are easier to identify. For example, in religious or grandiose delusions, people may believe they are holy or are the president of the country. Some delusions are quite bizarre. For example, people may think that their organs have all been replaced by machine parts or that their head contains a radio that receives messages from the government.

Disorganized speech

Disorganized speech refers to speech that does not contain the expected logical connections between thoughts or between questions and answers. For example, people may jump from one topic to another without ever finishing a thought. The topics may be slightly related or entirely unrelated. In other cases, people respond to simple questions with long, rambling answers, full of irrelevant detail. Answers may be illogical or completely incoherent. This type of speech differs from the difficulty expressing or understanding language ([aphasia](#)) or forming words ([dysarthria](#)) that is caused by a brain disorder such as stroke.

Occasionally misspeaking or intentionally being evasive, rude, or humorous is not considered disorganized speech.

Disorganized behavior

Disorganized behavior refers to doing quite unusual things (such as undressing or masturbating in public or shouting and swearing for no apparent reason). People with disorganized behavior typically have trouble doing normal daily activities (such as maintaining good personal hygiene or obtaining food).

Hallucinations

Hallucination refers to hearing, seeing, smelling, tasting, or feeling things that are not actually there. That is, people perceive things, seemingly through their senses, that are not caused by an outside stimulus. Any sense can be involved. The most common hallucinations involve hearing things (auditory hallucinations), usually voices. The voices often make derogatory comments about the person or command the person to do something.

Not all hallucinations are caused by a mental health condition. [Psychedelic drugs](#), such as LSD, mescaline, and psilocybin, are referred to as hallucinogens because they can cause visual hallucinations. Some types of hallucinations are more likely to be caused by a [neurologic disorder](#). For example, before a seizure occurs, people may smell something when there is no smell (an olfactory hallucination).

Mood extremes

Mood extremes include outbursts of rage, periods of extreme elation ([mania](#)) or [depression](#), and, conversely, constant expression of little or no emotion (appearing unresponsive or apathetic).

Causes of Personality and Behavior Changes

Although people sometimes assume that changes in personality, thinking, or behavior are all due to a mental illness, there are many possible causes. All causes ultimately involve the brain, but dividing them into 4 categories can be helpful:

- Psychological disorders
- Substance use (including drug intoxication or withdrawal) and medication side effects
- Disorders that affect mainly the brain

- Bodywide (systemic) disorders that also affect the brain

Mental illnesses

Mental illnesses include

- [Anxiety disorders](#)
- [Bipolar disorder](#)
- [Depression](#)
- [Dissociative disorders](#) (when people feel detached from themselves or from the world around them)
- [Obsessive compulsive disorders](#)
- [Personality disorders](#)
- [Schizophrenia](#) or other psychotic disorders
- [Somatic disorders](#) (when psychological factors are expressed as physical symptoms)
- [Trauma- and stressor-related disorders](#)

Medications and illicit/recreational drugs

Substances or medications may affect personality or behavior when they cause

- Intoxication: Particularly [alcohol](#) (when consumed in large amounts), [amphetamines](#), [cocaine](#), [hallucinogens](#) (such as LSD), and [phencyclidine](#) (PCP)
- Withdrawal: [Alcohol](#), [barbiturates](#), [benzodiazepines](#), and [opioids](#)
- [Side effects](#): Medications intended to affect brain function (including [antiseizure medications](#), [antidepressants](#), [antipsychotics](#), [sedatives](#), and stimulants), [medications with anticholinergic effects](#) (such as antihistamines), [opioid pain relievers](#), and corticosteroids

Rarely, certain antibiotics and high blood pressure medications cause changes in personality and behavior.

Disorders that mainly affect the brain

These brain (neurologic) disorders can affect personality, mood, and behavior. They include

- [Alzheimer disease](#)
- Brain infections, such as [meningitis](#), [encephalitis](#), and human immunodeficiency virus (HIV) infection that involves the brain (called [HIV-associated encephalopathy](#))

- [Brain tumors](#)
- Head injuries, such as a [concussion](#) and [postconcussion syndrome](#)
- [Multiple sclerosis](#)
- [Parkinson disease](#)
- [Seizure disorders](#)
- [Stroke](#)

Bodywide disorders that also affect the brain

Bodywide (systemic) disorders that also affect the brain include

- [Kidney failure](#)
- [Liver failure](#)
- Low blood sugar ([hypoglycemia](#))
- [Systemic lupus erythematosus](#) (lupus)
- Thyroid disorders, such as an underactive thyroid gland ([hypothyroidism](#)) or an overactive thyroid gland ([hyperthyroidism](#))

Less commonly, [Lyme disease](#), [sarcoidosis](#), [syphilis](#), or a vitamin deficiency causes personality and behavior changes.

Evaluation of Personality and Behavior Changes

During the initial evaluation, doctors try to determine whether symptoms are due to a mental illness or a general medical disorder.

The following information can help people decide when a doctor's evaluation is needed and help them know what to expect during the evaluation.

Warning signs

In people with changes in personality or behavior, certain symptoms and characteristics are cause for concern. These warning signs include

- Symptoms that appear suddenly
- Attempts to harm themselves or others or threats to do so
- Confusion or delirium
- Fever
- Severe headache

- Symptoms that suggest malfunction of the brain, such as difficulty walking, balancing, or speaking or vision problems
- A recent head injury (within several weeks)

When to see a doctor

People who have warning signs should be seen by a health care professional as soon as possible. Law enforcement may need to be called if people are violent.

What the doctor does

Doctors first ask questions about the person's symptoms and medical and psychiatric history. Doctors then do a physical examination, including a [neurologic examination](#) with a [mental status examination](#) (which evaluates the ability to pay attention, memory, mood, and ability to think abstractly, follow commands, and use language, among other things). What they find during the history and physical examination often suggests a possible cause of the changes and the tests that may need to be done (see table [Some Causes and Features of Personality and Behavior Changes](#))

Questions include when symptoms began. Many mental health conditions begin in a person's teens or 20s. If a mental health condition begins during middle age or later, especially if there is no obvious trigger (such as loss of a loved one), the cause is more likely to be a general medical disorder. A general medical disorder is also more likely to be the cause when psychological symptoms change significantly during middle age or later in people with a chronic mental health condition. If changes began recently and suddenly in people of any age, doctors ask about conditions that can trigger such changes. For example, they ask whether people have just started or stopped taking a prescription or a recreational or illicit drug.

Doctors ask about other symptoms that may suggest a cause, such as

- Palpitations: Possibly an overactive thyroid gland or use or withdrawal of a medication
- Tremors: Parkinson disease or withdrawal of a medication
- Difficulty walking or speaking: Multiple sclerosis, Parkinson disease, stroke, or intoxication from an opioid or a sedative
- Headache: Brain infection, brain tumor, or bleeding in the brain (hemorrhage)
- Numbness or tingling: A stroke, multiple sclerosis, or a vitamin deficiency

People are also asked whether they have previously been diagnosed and treated for a psychological or seizure disorder. If they have been treated, doctors ask whether they have stopped taking their medications or decreased the dose. However, because

people with mental health conditions may also develop physical conditions, doctors do not automatically assume that any new abnormal behavior is caused by the mental health condition.

Doctors ask about general medical disorders people have (such as diabetes) and about their lifestyle (such as their marital status, job situation, educational background, use of alcohol, tobacco, and recreational drugs, and living arrangements). Doctors also ask whether family members have had any physical disorders that can cause psychological symptoms (such as multiple sclerosis).

During the **physical examination**, doctors look for signs of general medical disorders that can cause changes in mental status, particularly the following:

- Fever and/or a rapid heart rate (suggesting an infection, alcohol withdrawal, or use of amphetamines or cocaine in high doses)
- Confusion or delirium (suggesting drug intoxication or withdrawal)
- Abnormalities during the neurologic examination, such as difficulty forming words or understanding language (possibly suggesting a brain disorder)

Confusion and delirium are more likely to result from a general medical condition. People with mental health conditions are rarely confused or delirious. However, many medical disorders that cause changes in behavior do not cause confusion or delirium, but they often cause other symptoms that may appear to be a mental health condition.

Doctors bend the person's neck forward. If doing so is difficult or painful, meningitis may be the cause. Doctors check the legs and abdomen for swelling, which may result from kidney or liver failure. If the skin or whites of the eyes look yellow, the cause may be liver failure.

Doctors may examine the interior of the eyes with a handheld device that looks like a small flashlight (called an ophthalmoscope). If doctors see swelling in part of the optic nerve (papilledema), pressure within the skull may be increased, and tumors or bleeding in the brain may be the cause of the mental symptoms.



Some Causes and Features of Personality and Behavior Changes

Cause	Common Features*	Tests
Mental disorders	Symptoms that occur in episodes lasting a few weeks to a few months and that include mania (abnormally elevated mood or energy level), depression, or both Episodes of mania: • Elation or irritability • Grandiosity (exaggerated sense of superiority, power, knowledge, or identity) • Talkativeness • Increased activity • A decreased need for sleep • Racing thoughts, jumping from one idea to another • Sometimes hallucinations or delusions of persecution	A doctor's examination
<u>Bipolar disorder</u>		

